

Boilertronics

Sprint 2 Planning Document

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Sprint Overview

This sprint we are focused on building the level creator infrastructure and improving upon prior frameworks for various level mechanisms. The game should ideally look cleaner and be easier to navigate. The game should also have more depth to some of the mechanics developed in the first sprint. This will allow us to have an intuitive game that is easy to pick up.

Scrum Master: Ethan Dixon

Meeting Plan:

- Tuesday, 6:30 PM
- Thursday, 6:00 PM with Project Coordinator and work time after

Risks and Challenges

This sprint's development work directly builds on top of the systems we developed in the prior sprint. There is a degree of uncertainty and some challenge that features and enhancements may be bogged down by sloppier design choices from the prior sprint. This should in theory not be an issue, but there is some risk that we may see emergent issues related to this. We see more interconnectedness between our systems, but less direct dependencies.

Another risk is based on the complexity and size of the system. As we implement more systems and subsystems, we need to support more interactions between different elements. While the baseline framework appears to be stable, the increase in moving parts runs the risk of hitting edge cases that previously were not considered or planned around. In the event of errors or unintended behaviors, it may become more difficult to diagnose the root cause of the issue.

Current Sprint Detail

User Story #5

As a developer, I want to have a level creator framework ensuring additional levels can be created, with an interface to create levels.

#	Description	Estimated Time	Owner
1	Create UI elements that provide a more user-friendly way to work with "protected tiles",	3 hrs	Ethan D

	including a toggle for visibility of said tiles, a proper menu/interface to toggle each tile's "protected" states.		
2	Create a UI element that allows the developer to change the metadata of the level (name, size, etc.).	2 hrs	Ethan D
3	Create a custom menu to place objects that are otherwise not placeable by the user (i.e. level-specific input/output objects, obstructions)	1 hr	Ethan D
4	Create a basic drop-down menu that will let the developer select the background tiles (i.e. outside of the play area) for a level.	2 hrs	Ethan D
5	Create a special save menu that allows for the creation of new levels and the overriding of old levels.	2 hrs	Ethan D
6	Implement a way to select different levels on the level select screen.	1 hr	Ethan D
7	Implement a way to show the cost of an item in the placeable items menu.	1 hr	Ethan D
8	Implement test cases to ensure functionality.	2 hrs	Ethan D

Acceptance Criteria:

- Given the developer is interacting with the level creator, if they click a tile, then a menu will pop up allowing them to change the tile data.
- Given the developer is interacting with the level creator, if they want to change the metadata of the level, then they will be able to through UI elements.
- Given the developer is interacting with the level creator, if they want to place a developer specific item, then they will be able to place it through a UI element.
- Given the developer is interacting with the level creator, if they want to change the background tiles, then they will be able to decide through a UI element.
- Given the developer is interacting with the level creator, if they want to save new levels, then there will be a level UI element to do so.
- Given the developer is interacting with the level select menu, if they want to load a specific level, then they will be able to.
- Given the developer is interacting with the level creator, if the placeable items menu is open, it will display the cost of each item.

User Story #6

As a user, I would like a leaderboard system to compare myself to my previous scores.

#	Description	Estimated Time	Owner
1	The leaderboard should maintain the top score for each category, updating if the last completed solution is an improvement in any of the statistical categories	2 hrs	Abhi
2	Create a custom leaderboard save system to save and load the score of a user. For now, this will save to a local location – this will ideally point to some server in Sprint 3.	3 hrs	Ethen C.
3	Leaderboard system should display the data	1	Zach

Acceptance Criteria:

- Given that the user completes a level with a more optimal solution, the leaderboard should update the statistics if the current value is less optimal than in the recent solution
- Given that the user accesses the leaderboard menu, they should be able to view saved leaderboard statistics.
- Given that the user access the leaderboard menu, the saved leaderboard statistics should be displayed for all locally saved slots

User Story #7

As a user, I want to be able to configure the time each 'step' takes in running my solution to the level (speed up / slow down / pause).

#	Description	Estimated Time	Owner
1	Configure the current movement system to use different time steps (move at different rates)	2 hrs	Keenan
2	Create run functions that will move elements at different speeds automatically. (1x and 2x speed)	2 hrs	Keenan

3	Create submit solution function that will slowly ramp up time until an error is encountered, or the level is complete.	3 hrs	Keenan
4	Create a level to demonstrate these features and write a set of user interactions for testing	1 hr	Keenan

Acceptance Criteria:

- Given the user is running their solution, if their solution encounters an error, it will stop running,
- Given the user is running their solution, if the user presses pause, it will stop running,
- Given the user is running their solution, if their solution completes the goal for the level, it will stop running,
- Given the user is running their solution, the user can run their solution at 1x speed.
- Given the user is running their solution, the user can run their solution at 2x speed.
- Given the user is running their solution, the user can 'submit' their solution, which will slowly ramp up the speed the solution runs at until one of the conditions above is met and stops the solution.

User Story #8 - Continuation from last Sprint (Only Red)

As a user, I want to stop the level if an error or collision occurs in the user solution.

#	Description	Estimated Time	Owner
1	In the event of an element collision, a user cannot step the program until they reset the level	0 hrs	Abhi
2	In the event of a program error, the code cannot be stepped forward	0 hrs	Abhi
3	In the event of an element going out of bounds the user	0 hrs	Abhi

	cannot step the program until they reset the level		
4	The program should identify when a movable element goes out of bounds or collides with another element	2 hr	Abhi
5	Create unit tests for each type of error	0 hrs	Abhi

Acceptance Criteria:

- Given that the user provides programmatic instructions that lead to a collision where 2 or more of the same elements are located in the same x,y position after the step, a collision error should be noted to the user
- Given that the user provides programmatic instructions that lead to a collision where 2 or more of the same elements are moved into each other such that their trajectories attempt to move past each other, a collision error should be noted to the user
- If two objects are moving such that one moves to a location where another previously existed, no error should occur

User Story #10 - Continuation from last Sprint (Only Red)

As a user, I want to see when my solution encounters an error.

#	Description	Estimated Time	Owner
1	Developing specific error popups (vary by error)	0 hrs	Abhi
2	Enabling popups to initiate automatically from the error	0 hrs	Abhi
3	Stopping the code terminal highlighted step on the error producing line, distinguishing it from the normal display	0 hrs	Abhi

4	Displaying the error/collision within the running level UI for each type of error, no matter where in the solution it occurs	1.5 hr	Abhi
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Acceptance Criteria:

- Given that the user is interacting with a game level, then the error within the solution itself will be depicted in a manner that distinguishes it from the rest of the level ui no matter where in the level it occurs
- Given the user solution experiences a terminal error (syntax related) the level ui will distinguish the error in a manner which is unique from a typical logic error
- If multiple errors occur, syntax and object related, the syntax error takes priority with its line highlighted red but the object error popup should open as well

User Story #15

As a user, I would like to be graded on my solution's statistics such as time to finish, total resources consumed, resources produced per second, and other such factors.

#	Description	Estimated Time	Owner
1	Add sections to level statistics ui	2 hr	Ethan D
2	Set up framework for how statistics will be stored/saved and link to ui (unique identifier for variables and how they should update)	1 hr	Abhi
3	Add logic for tracking interactions which impact statistics, within each function where the interaction occurs (incrementing cost when added and decrementing when removed)	3 hrs	Abhi
4	The cost variable will automatically adjust based on items being added or removed	2 hrs	Abhi
5	Ensure that end level statistics are not shown/printed for erroneous solutions or solutions which	2 hrs	Abhi

	do not complete the level and are printed only upon successful level completion		
6	The reset button should reset the level statistics to default values and difference calculations	1 hr	Abhi
7	Create test cases varied by efficiency to test whether grading system is reasonably correct	1 hr	Abhi
8	Give user a calculated grade based on difference between user solution and cutoff	1 hr	Abhi

Acceptance Criteria:

- Given the user is playing a level, if they press the level statistics button, the data will be divided into different sections.
- Given the user places additional items onto the field, the cost counter increases accordingly based on the element's associated cost
- Given the user removes items from the field, the cost counter decreases accordingly based on the element's associated cost
- Given the user has not yet reached a completed solution, the statistics menu does not show statistics (or displays N/A or 0) for the user solution statistics
- Given the user completes a level, they can see a computed grade on various statistics based on their computed values against the static cutoff values
- Given the user clicks the reset button, the solution statistics are deleted/reset back to placeholder values (N/A or 0)

User Story #16

As a user, I would like to be given grading “cutoffs” to have a point of reference for ideal solutions.

#	Description	Estimated Time	Owner
1	Create various test cases for potential solutions to serve as a guide for the cutoffs	1 hrs	Abhi
2	Add the grading cutoffs as static values within the Statistics menu	1 hr	Abhi

3	Allow for user to see differences in solution statistics vs the cutoff presented as being above or below the threshold and demonstrated as being good/bad based on color/highlighting (dynamic based on solution)	3 hrs	Abhi
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Acceptance Criteria:

- Given the user opens the level statistics ui they will have static “grading cutoff” values for the grading categories
- Given the user completes the level, they will be able to see their statistics for the completed level to compare with the grading cutoff
- Given the user completes the level, they can see the computed difference between their solution statistics and the cutoff values to understand how “far off” their solution is

User Story #17

As a user, I would like to be presented with unchangeable obstructions, including (but not limited to) such obstructions such as missing tiles on any layer.

#	Description	Estimated Time	Owner
1	Create floor obstruction sprite(s) (cracked tiles, pipes) for obstruction objects.	0.5hrs	Ethen Chen
2	Create ceiling obstruction sprite(s) (stalagmites, pipes) for obstruction objects.	0.5hrs	Ethen Chen

Acceptance Criteria:

- Given that the user is interacting with a playable level, floor obstruction objects may exist if the level requires such objects to exist.
- Given that the user is interacting with a playable level, ceiling obstruction objects may exist if the level requires such objects to exist.
- Given that the user is interacting with a playable level, obstruction objects must be “protected” tiles.

User Story #23

As a developer, I would like the validity of the script to be checked on runtime, on a line by line basis.

#	Description	Estimated Time	Owner
1	Error checking should occur when the run button is pressed across the entire terminal	2 hrs	Zach
2	If any major syntax error exists in the code, the script will not run at all	2 hrs	Zach
3	Modify syntax error warning to list all problematic lines in a terminal	2 hrs	Zach
4	Write Test Document	1 hr	Zach

Acceptance Criteria:

- Assuming the user is in a level, pressing the run button should generate errors on all invalid lines of the terminal
- No code should be executed if any error exists across all terminals
- When multiple errors are present, the bottom of terminal error bar will display all problematic lines and the first occurring error message
- All errors present in a terminal will be marked as incorrect
- All errors across all terminals will be marked as incorrect

User Story #24

As a developer, I would like there to be dynamic error checking in the code terminal.

#	Description	Estimated Time	Owner
1	Incorrect Syntax highlighting should occur for invalid code when a user starts a new line	1 hr	Zach
2	Incorrect Syntax highlighting should occur for invalid code when a user selects a new line	1 hr	Zach
3	Incorrect Syntax	1 hr	Zach

	highlighting should occur for invalid code when a user selects a different terminal		
4	Implement a syntax pre-processor for dynamic error checking prior to interpretation	2 hrs	Zach
5	Develop test cases	1 hr	Zach

Acceptance Criteria:

- Assuming the user is in a level and entering values into the terminal, pressing the enter key while having a syntactically incorrect command entered will show a syntax error, syntax highlighting
- Assuming the user is in a level and entering values into the terminal, clicking onto a new line while having a syntactically incorrect command entered will show a syntax error, syntax highlighting
- Assuming the user is in a level and entering values into the terminal, clicking onto a new tab while having a syntactically incorrect command entered will show a syntax error, syntax highlighting

User Story #26

As a developer, I would like to further extend the instruction set to support more registers and the concept of “null”.

#	Description	Estimated Time	Owner
1	Implement conditional jumps like jne, jge	3 hrs	Zach
2	Implement a visual display for the registers visible to the user	2.5 hrs	Zach
3	Implement flag tied to terminal/object allowing for variable amounts of registers	1.5 hrs	Zach

4	Implement register decay feature such that register memory is volatile and the value will be overwritten after a fixed number of steps, on expiration it is null	2 hrs	Zach
5	Implement automated test case framework for terminal specific testing	2 hrs	Zach
6	Automated test cases	1	Zach

Acceptance Criteria:

- Given the user is in a level and writing valid code in a terminal, the je command jumps to the specified point if the comparison register indicates it should
- Given the user is in a level and writing valid code in a terminal, the jne command jumps to the specified point if the comparison register indicates it should
- Given the user is in a level and writing valid code in a terminal, the jge command jumps to the specified point if the comparison register indicates it should
- Given the user is in a level and writing valid code in a terminal, the jg command jumps to the specified point if the comparison register indicates it should
- Given the user is in a level and writing valid code in a terminal, the jl command jumps to the specified point if the comparison register indicates it should
- Given the user is in a level and writing valid code in a terminal, the jle command jumps to the specified point if the comparison register indicates it should
- Given the user is in a level and writing valid code in a terminal, there is a visual representation for the registers in use that changes when the register values are changed
- Given the user is in a level and writing valid code in a terminal, if a register is not used for 5 steps, the value will be overwritten to null
- Performing a comparison or arithmetic operation with null will generate an error

User Story #27

As a user, I would like there to be a technical manual on the in game programming language.

#	Description	Estimated Time	Owner
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1	Document the syntax, semantics, usage, and relevant mechanisms for the user to be used in hints/reference/manual menu	3 hr	Zach
2	Ensure manual is viable in game from settings with popup	1 hrs	Zach
3	Develop popup for manual that is accessible from any given level	1 hrs	Zach
4	Reference segments linked to reference menu in levels	.5	Zach
5	Test doc for ensuring all popups appear correctly	.5	Zach

Acceptance Criteria:

- There must exist a manual detailing the assembly language, syntax, usage, and mechanics in detail
- Given the user is on the main menu, if the player clicks through settings then there will be an option for the manual
- Given the user is in a level, if the player opens the reference menu then there will be an option for a subset of the manual

User Story #29

As a user, I would like to have the scripts run synchronously such that all the steps happen at the same time.

#	Description	Estimated Time	Owner
1	Allow for objects to move synchronously during execution of the solution	2 hrs	Keenan
2	Stress test the synchronous movement system with both a large amount of objects and varying speeds of objects moving. Make	2 hrs	Keenan

	adjustments to the movement system to improve performance if needed.		
3	Add checks to make sure that the next step does not execute until all the objects are finished moving/executing	2 hrs	Keenan
4	Create a test level to demonstrate synchronous execution of scripts and synchronous movement of objects	1 hr	Keenan

Acceptance Criteria:

- Given the user is stepping through their solution, all claws should step at the same time.
- Given the user is stepping through their solution, all conveyors should step at the same time.
- Given the user is stepping through their solution, claws and conveyors should step at the same time.
- Given the user is stepping through their solution, the user should not be able to step until the objects stop moving.

User Story #31

As a user, I would like a quick reference panel / menu within levels for global hints based on progress.

#	Description	Estimated Time	Owner
1	Implement a UI element that shows the user hints in the level.	2 hrs	Ethan D
2	Generate terminal related hints based on level state values	1 hr	Zach
3	Implement test cases	.5 hr	Ethan D

Acceptance Criteria:

- Given the user is in the level playspace, if they press the designated hint button, then there is a popup that will contain hints.
- Given the user is in the level space, if the user clicks an arrow in the hint popup, then there will be displayed a new hint
- Given the user is in the level space, if the user clicks an arrow in the hint popup, then all existing hints can be cycled through

User Story #35

As a user, I want the floor layer to have default visuals and each of the placeable elements that can be put on the floor layer to have separate default visuals too.

#	Description	Estimated Time	Owner
1	Draw up alternative floor tiles	0.5hrs	Ethen C
2	Draw up alternative factory sprites	0.5hrs	Ethen C
3	Update PlaceableObject.cs and systems (including Layer.cs) to support objects with variable dimensions, i.e. 1x2, 2x3, etc. This includes how variable-dimension objects may be rotatable!	2hrs	Ethen C
4	Update Moveable, Scriptable, and other such systems to support interactions with these variable dimension objects	2hrs	Keenan
5	Implement some UI system that allows for these variable dimension objects to be rotated	2hrs	Ethan D
6	Update object dragging visuals to account for these variable dimension objects, including considerations for their directions, etc.	2hrs	Keenan
7	Make the new factory elements (furnace, press, and other machines), Runnable elements, and make them be able to give an object by a claw and allow a claw to take an object from them when they are done with their operation.	3 hrs	Keenan
8	Create a test level and instructions to demonstrate the functionality of the new factory layer objects	1 hr	Keenan

Acceptance Criteria:

- Given that the user is in a level, they should be able to click and drag to place factory layer machines that take up multiple tiles

- Given that the user is in a level, they should be able to move factory layer machines that take up multiple tiles
- Given that the user is in a level, they should be able to delete factory layer machines that take up multiple tiles
- Given that the user is in a level, if the user wants to rotate a variable dimension object, they are able to do so through a UI element.
- Given that the user is running their solution, a factory layer machine should be able to receive a resource via a claw.
- Given that the user is running their solution, a factory layer machine should be able to have a claw grab a resource from it.

User Story #36

As a user, I want each placeable element on the movement layer to have visuals beyond a baseline placeholder sprite.

#	Description	Estimated Time	Owner
1	Draw up basic "switch" variable dimension object sprites (2x1)	0.5 hrs	Ethen C
2	Update Moveable, Scriptable, and other such systems to support these Moveable variable dimension objects	2hrs	Keenan
3	Implement the switch object as both a runnable and scriptable object and allow for desired functionality	2 hrs	Keenan
4	Add the 'switch' command to the assembly language	1 hr	Keenan
5	Create test level and instructions to demonstrate the functionality of the new movement objects	1 hr	Keenan

Acceptance Criteria:

- Given that the user is interacting with a playable level, a "switch" (an object of 2x1 size) must be placeable/must be placed on the level.
- Given that the user is interacting with a playable level, the "switch" object must be programmable.
- Given that the user is interacting with a playable level, the "switch" object must properly "swap" the position of two claws.

User Story #37

#	Description	Estimated Time	Owner
1	Draw up basic “grab” sprites for claws of various materials	0.5 hrs	Ethen C
2	Update PlaceableObject and systems to support objects with alternative sprites; avoid altering the save system if possible.	2hrs	Ethen C
3	Update Moveable and systems to support moving objects with alternative sprites	2hrs	Keenan
4	Fully implement the drop and grab command so that claws can interact with elements on the factory layer.	2 hrs	Keenan
5	Create level and instruction set to demonstrate the functionality of the new claw objects	0.5 hrs	Keenan

Acceptance Criteria:

- The “claw” object must have alternative states for when it is picking up an object.
- The “claw” object, when picking up an object, must visually show the corresponding “picked up item” as “grabbed” within the claw.
- The “claw” object must have its correct grabbing sprite when the “claw” is moving due to user programmed code.
- The “claw” object must be able to grab items from the floor and from machines on the factory layer.
- The “claw” object must be able to drop items onto the floor or into a machine on the factory layer

User Story #40

As a user, I want to toggle the view of a specific layer on and off.

#	Description	Estimated Time	Owner
1	Allow for the visibility of the floor layer to be toggled	2 hrs	Ethan D
2	Allow for the visibility of the claw layer to be toggled	2 hrs	Ethan D

3	Allow for the visibility of the movement layer to be toggled	2 hrs	Ethan D
4	Create a menu that contains buttons for each layer's visibility settings	1 hr	Ethan D
5	Create test cases to ensure functionality	1 hr	Ethan D

Acceptance Criteria:

- Given the user is in the level UI playspace, if they toggle the floor visibility button, the floor layer should disappear and reappear accordingly.
- Given the user is in the level UI playspace, if they toggle the floor visibility button, the floor layer should disappear and reappear accordingly.
- Given the user is in the level UI playspace, if they toggle the floor visibility button, the floor layer should disappear and reappear accordingly.
- Given the user is in the level UI playspace, if they click the visibility button, then there is a menu to toggle layer visibilities.

User Story #41

As a user, as I am placing tiles, I want the tiles on the corresponding layer to be more opaque than others for visibility improvements.

#	Description	Estimated Time	Owner
1	When placing on a specific tile on the floor layer, that layer will be more opaque than the other layers.	2 hrs	Ethan D
2	When placing on a specific tile on the movement layer, that layer will be more opaque than the other layers.	2 hrs	Ethan D
3	When placing on a specific tile on the claw layer, that layer will be more opaque than the other layers.	2 hrs	Ethan D
4	Create test cases to ensure functionality	1 hr	Ethan D

Acceptance Criteria:

- Given a user is interacting with the level playspace, if they select an item from the floor items menu and hover it over the floor layer, then the other layers will become more transparent.

- Given a user is interacting with the level playspace, if they select an item from the movement items menu and hover it over the movement layer, then the other layers will become more transparent.
- Given a user is interacting with the level playspace, if they select an item from the claw items menu and hover it over the claw layer, then the other layers will become more transparent.
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User Story #44

As a user, I would like sound effects to accompany actions from interactable game mechanisms, time iteration, and other mechanics.

#	Description	Estimated Time	Owner
1	Obtain sound effects from online resources for each interaction within the level	1 hr	Abhi
2	Attach sounds to the project and adjust as necessary within Godot system	1 hr	Abhi
3	Attach sounds to each type of individual interaction whether object related, terminal related, error related	3 hrs	Abhi
4	Cut off all sounds when reset button pressed	1 hr	Abhi
5	Create test cases for each interaction to ensure sounds operate as expected	1 hr	Abhi

Acceptance Criteria:

- Given that an object moves or is updated by the user code, a sound should emit
- Given that an object performs an action, its sounds should be differentiable from other objects if the action performed is unique (i.e. claws grabbing should not sound like conveyors moving but conveyors moving should sound similar to claws moving)
- Given that an error is encountered, a unique error sound should play

User Story #45

As a user, I want to be able to change the game's volume settings.

#	Description	Estimated Time	Owner
1	Implement a volume slider and mute button that allows the user to change the audio of the game.	2 hrs	Abhi
2	Implement test cases to ensure correct volume loudness.	.5 hr	Abhi

Acceptance Criteria:

- Given the user is in the Level UI settings menu, if they move the volume slider, then the volume will adjust accordingly.
- Given the user is in the Level UI settings menu, if they interact with the mute button, it should mute the audio.
- Given the user is in the main settings menu, if they move the volume slider, then the volume will adjust accordingly.
- Given the user is in the main settings menu, if they interact with the mute button, it should mute the audio.

User Story #52

As a developer, I want a level editing system to make designing new levels easier.

#	Description	Estimated Time	Owner
1	Implement some toggle that enables "level creation" mode within the main level scene. If we were instead to copy the level functionality to a separate scene, this could result in potential "desync" between the level creator and the playable level scenes as features get developed. Alternatively, look into actually creating a secondary scene that largely copies the functionality of the base playable level scene, and be sure to maintain this scene accordingly to keep up with updates with the main scene.	4 hrs	Ethen C

2	Create a rendering system (for debugging/level creation purposes) that will highlight “protected” tiles	2hrs	Ethen C
3	Create some setting and system that allows for tiles to be earmarked as “protected” or not, whether this is via hacking the tile placement system, or just by using keybinds (for now).	3hrs	Ethen C
4	Update the save system to allow for saving directly to levels – use a new function to differentiate from the standard save functionality!	2hrs	Ethen C
5	Update the save system to account for potential new types of metadata, i.e. restricted placeable objects, etc	3hrs	Ethen C
6	Implement automatic floor generating system that should ideally “fill” the outside of the placement area with desaturated, non-interactable floor tiles; also likely include a metadata option for saves to determine what tile to use to fill (but only as an image – memory cost for X amount of PlaceableObjects that exist only as background filling is likely too high)	3hrs	Ethen C
7	Update camera system to only allow panning to a certain point past the placement area	1hrs	Ethen C
8	Write up testing documentation for level creation functionality	2hrs	Ethen C

Acceptance Criteria:

- Given that the user is a developer, a developer “level creation” scene must be accessible by the developers.
- Given that the developer is interacting with this “level creation” scene, this scene must allow for developers to mark specific tiles as “protected” or not.
- Given that the developer is interacting with this “level creation” scene, this scene must correctly create (or overwrite) existing base levels, saving them to file.
- Given that the user is interacting with a “play” scene, levels should have automatically generated background tiles to fill up the visual “dead space” of a level

Total Hours Per Person for Sprint 2:

Name	Total Time
Abhi	31 hrs
Ethan D	35.5 hrs

Ethen C	30 hrs
Keenan	33.5 hrs
Zach	32 hrs

Remaining Backlog: (2025/10/21)

Functional Requirements:

- ~~1.) As a user, I want there to be different movable, placeable elements (i.e. claws, rails, movement tiles, machines) to place on different types of gridded layers (factory layer, claw layer, and movement layer). Additionally, I want to make sure that I can only place elements where and when permitted.~~
- ~~2.) As a developer, I want each "step" to allow the programmable objects to move.~~
- ~~3.) As a developer, I want the view of the game/tiles to be isometric.~~
- 4.) As a user, I would like an overarching story via a text block to better engage the user and to provide context between groups of levels.
- ~~5.) As a developer, I want to have a level creator framework ensuring additional levels can be created, with an interface to create levels.~~
- ~~6.) As a user, I would like a leaderboard system to compare myself to my previous scores.~~

- Level Functionality

- ~~7.) As a user, I want to be able to configure the time each 'step' takes in running my solution to the level (speed up / slow down / pause).~~
- ~~8.) As a user, I want to stop the level if an error or collision occurs in the user solution.~~
- ~~9.) As a user, I want to be able to run my solution step by step.~~
- ~~10.) As a user, I want to see when my solution encounters an error.~~
- ~~11.) As a developer, I want to require the user to reset their solution before being able to change their solution.~~
- ~~12.) As a user, I want to be able to reset my solution at any point while I'm stepping through my solution.~~
- ~~13.) As a user, I want to be able to save and load my solutions.~~
- 14.) As a developer, I would like a mechanism that delineates the "quality" of different interactable components in the level that have varying cost/performance factors based on their operation cost.
- ~~15.) As a user, I would like to be graded on my solution's statistics such as time to finish, total resources consumed, resources produced per second, and other such factors.~~
- ~~16.) As a user, I would like to be given grading "cutoffs" to have a point of reference for ideal solutions.~~
- ~~17.) As a user, I would like to be presented with unchangeable obstructions, including (but not limited to) such obstructions such as missing tiles on any layer.~~

- Programming Language/Editor

- ~~18.) As a user, I want to have multiple sidebar terminal windows where I can write my code for the level.~~

- 19.) — ~~As a user, I would like to be able to write code in the in-game code editor which will be interpreted into function calls to control factory elements.~~
- 20.) — ~~As a user, I want a visual indicator that the code I have written is either correct or has some form of syntax error.~~
- 21.) — ~~As a developer, I want the programming language to support stoppable commands like “grab,” “wait,” “move.”~~
- 22.) — ~~As a developer, I would like the programming language to interact with defined, scriptable objects to interact with the level in an iterative, step-based process.~~
- 23.) — ~~As a developer, I would like the validity of the script to be checked on runtime, on a line by line basis.~~
- 24.) — ~~As a developer, I would like there to be dynamic error checking in the code terminal, if time allows.~~
- 25.) — ~~As a user, I want the code editor to have different “tabs” corresponding to different elements within the level.~~
- 26.) — ~~As a developer, I would like to further extend the instruction set to support more registers and the concept of “null” if time allows.~~
- 27.) — ~~As a user, I would like there to be a technical manual on the in-game programming language.~~
- 28.) As a developer, I would like there to be a gradual reveal of information pertaining to the full range and scope of the programming language to the user over the course of level progression.
- 29.) — ~~As a user, I would like to have the scripts run synchronously such that all the steps happen at the same time.~~

- UI/UX

- 30.) — ~~As a user, I want to be able to navigate through the game via menus: main to settings, main to level sections, level sections to levels.~~
- 31.) — ~~As a user, I would like a quick reference panel / menu within levels for global hints based on progress.~~
- 32.) As a user, I would like a hint system that would allow the user to see the quick reference for the concept introduced in the level.
- 33.) — ~~As a user, I want the level UI to be divided into different sections (Statistics, Placeable Items, Exit Menu, Time Interaction [Pause, play], etc).~~
- 34.) — ~~As a user, I want to be able to move and re-order my code editor tabs.~~
- 35.) — ~~As a user, I want the floor layer to have default visuals and each of the placeable elements that can be put on the floor layer to have separate default visuals too.~~
- 36.) — ~~As a user, I want each placeable element on the movement layer to have visuals beyond a baseline placeholder sprite.~~
- 37.) — ~~As a user, I want each placeable element on the claw layer to have visuals beyond a baseline placeholder sprite.~~
- 38.) — ~~As a user, I want to be able to separate between the different layers in an isometric view.~~

- ~~39.) As a user, I want to have the placement grid be aligned with the layer that the user is interacting with.~~
- ~~40.) As a user, I want to toggle the view of a specific layer on and off.~~
- ~~41.) As a user, as I am placing tiles, I want the tiles on the corresponding layer to be more opaque than others for visibility improvements.~~
- 42.) As a user, I want some sprites (claws moving up/down/left/right, machines working/creating items, and movement layers operating) to be animated when that element performs some action to give the user visual feedback.
- ~~43.) As a developer, I want trains to connect the levels in the level select menu.~~
- ~~44.) As a user, I would like sound effects to accompany actions from interactable game mechanisms, time iteration, and other mechanics.~~
- ~~45.) As a user, I want to be able to change the game's volume settings.~~

- Server Integration

- 46.) If time allows, as a user, I want to be able to create and login to an account.
- 47.) If time allows, as a user, I want to be able to play the game on Steam
- 48.) If time allows, as a user, I would like to be able to see other people's scores online and compare my score to theirs.
- 49.) If time allows, as a user, I would like to view the score my friends got online and compare my scores to theirs.
- 50.) If time allows, as a user, I would like to design my own levels and share them online.

- Extras

- 51.) As a user, I would like an achievements system derived from game and challenge completion.
- ~~52.) As a developer, I want a level editing system to make designing new levels easier.~~
- 53.) If time allows, as a developer, I would like to include easter eggs into the game.
- 54.) If time allows, as a user, I would like a block mode for the scripts rather than a strictly terminal based approach.
- 55.) If time allows, as a user, I would like to have a modding framework.
- 56.) If time allows, as a user, I would like more levels with different gimmicks such as pipes, electricity, heat, and other similar mechanics.
- 57.) If time allows, as a user, I would like to have colorblind modes.

Non-functional Requirements:

- 58.) As a user, I want good-looking tile sprites.
- 59.) As a user, I want responsive and intuitive buttons.
- 60.) As a user, I would like the game's visuals to be appealing and easy to read.
- 61.) As a user, I want to be able to scale the UI to make it easier to read.
- 62.) As a user, I want the animation for the given action I programmed to execute within 5 seconds of the game indicating that it should be executing. (performance)

- 63.) If time allows, as a user, I would like my data to be handled securely over the network when storing their solution metrics by utilizing HTTPS. (security)
- 64.) If time allows, as a developer, I want the user's data to be secure and not accessible by unauthorized parties.

Gantt Chart Below

[illegible]

[illegible]

[illegible]